

CONTRACT TITLE:
Taiwan Advanced Semiconductor Fab EPC Contract

CONTRACT VALUE (USD):
15000000000

JURISDICTION:
Taiwan

PROJECT LOCATION:
Hsinchu, Taiwan

SUPPLIER LOCATIONS (comma-separated):
Netherlands, Japan, USA, South Korea, Germany, France, Switzerland, Norway

 CONTRACT TEXT (paste into large text area):

ENGINEERING, PROCUREMENT AND CONSTRUCTION CONTRACT - TAIWAN ADVANCED
SEMICONDUCTOR MANUFACTURING FACILITY

This Agreement dated January 15, 2024 is between Taiwan Semiconductor Innovation Corp (Owner) and GlobalTech Construction Consortium led by Nordic Engineering AS (Contractor) for construction of a state-of-the-art 3nm semiconductor fabrication facility at Hsinchu Science Park, Taiwan.

CONTRACT VALUE: USD 1,500,000,000 (One Billion Five Hundred Million Dollars)
DURATION: 36 months (Completion: December 31, 2026)
JURISDICTION: Taiwan with ICC Arbitration in Singapore

SCOPE OF WORK:

The Contractor shall design, procure, construct, install, test and commission a 150,000 sq meter semiconductor fabrication facility including:

- 50 ASML EUV lithography machines (Netherlands)
- 200 Tokyo Electron etching systems (Japan)
- Applied Materials deposition tools (USA)
- Samsung Electronics integration systems (South Korea)
- Clean room facilities (Class 1 ISO standard)
- Ultra-pure water treatment plant (100,000 m³/day capacity)
- Electrical substation (500 MW capacity)
- Chemical storage and distribution systems
- Data center for fab automation (10 MW)

CRITICAL SUPPLY CHAIN DEPENDENCIES:

- ASML (Netherlands): EUV lithography - Route: Rotterdam → Suez Canal → Taiwan Strait → Kaohsiung Port
- Tokyo Electron (Japan): Etching equipment - Route: Yokohama → East China Sea → Taiwan Strait
- Applied Materials (USA): Deposition tools - Route: San Francisco → Pacific → Taiwan Strait
- Samsung (South Korea): Memory systems - Route: Busan → Yellow Sea → Taiwan Strait
- BASF/Merck (Germany): Electronic chemicals - Route: Hamburg → Suez → Taiwan Strait
- Air Liquide/Linde (France/Germany): Specialty gases - Route: European ports → Suez → Taiwan Strait

ALL EQUIPMENT MUST TRANSIT TAIWAN STRAIT - CRITICAL CHOKEPOINT

MATERIALS SUPPLIERS:

- Silicon wafers: Shin-Etsu Chemical (Japan), GlobalWafers (Taiwan)
- Photoresist chemicals: JSR Corporation (Japan), DuPont (USA)
- Structural steel: POSCO (South Korea), Nippon Steel (Japan)
- HVAC systems: Carrier (USA), Daikin (Japan)
- Electrical systems: Siemens (Germany), Schneider Electric (France)
- Precision optics: Zeiss (Germany), Nikon (Japan)

PAYMENT TERMS:

- Equipment procurement: USD 900M (60%)
- Construction: USD 400M (27%)
- Commissioning: USD 150M (10%)
- Contingency: USD 50M (3%)
- Monthly progress payments based on milestones
- 10% advance payment upon signing
- Liquidated damages: USD 500,000 per day for delays (max 10% of contract value)

PROJECT CRITICAL PATH:

- ASML EUV machine delivery Months 24-28 (CANNOT BE EXPEDITED)
- Taiwan Strait shipping window: Year-round dependency
- Clean room certification: Environmental controls
- Taiwan electrical grid connection: 500 MW load

FORCE MAJEURE CLAUSE:

Either party shall be excused from performance to the extent prevented by Force Majeure events including:

(a) Natural Disasters: earthquakes exceeding magnitude 6.0, typhoons, floods, tsunamis, volcanic eruptions typical for Taiwan Pacific Ring of Fire location

(b) Pandemic and Epidemic: WHO-declared Public Health Emergency of International Concern (PHEIC) including COVID-19, SARS, MERS, avian influenza, resulting in government-mandated lockdowns or international travel restrictions affecting personnel or cargo

(c) War and Military Conflict: declared war, invasion, or military blockade affecting Taiwan Strait, South China Sea, or Suez Canal shipping routes. Includes kinetic military action by state actors.

(d) Government Actions: expropriation, embargo, import/export restrictions, or sanctions by UN Security Council, US OFAC, or European Union affecting critical semiconductor equipment or materials

(e) Infrastructure Failure: prolonged electrical power grid failure exceeding 72 continuous hours, telecommunications network total collapse, or port closures exceeding 30 consecutive days

EXCLUSIONS FROM FORCE MAJEURE (Contractor bears risk):

- Economic hardship, cost increases, currency fluctuations, or inflation
- Labor disputes, strikes, work stoppages (unless government-mandated)
- Supplier delays due to commercial disputes, insolvency, or bankruptcy
- Cyber attacks, ransomware, or data breaches (unless proven state-sponsored)
- Routine weather conditions typical for Taiwan region
- Political tensions or diplomatic disputes without kinetic military action
- Terrorism, sabotage, or non-state actor attacks (NOT COVERED)
- Trade restrictions or tariffs by individual countries (NOT COVERED)
- Taiwan Strait tensions without declared war or blockade (NOT COVERED)
- Supply chain disruptions from factory shutdowns in supplier countries (NOT COVERED)
- Shipping delays from port congestion, vessel shortages, container crisis (NOT COVERED)
- Energy shortages, fuel price spikes, or electricity rationing (NOT COVERED)
- Airspace closures or flight restrictions (NOT COVERED)
- Satellite disruptions affecting GPS or communications (NOT COVERED)
- Commodity shocks for semiconductor materials, gases, chemicals (NOT COVERED)

- Technology export control changes (NOT COVERED - but critical risk)

NOTICE REQUIREMENTS:

- Affected party must provide written notice within 7 days of FM event
- Monthly progress updates required during FM continuation
- Failure to provide timely notice may void FM protection

MITIGATION OBLIGATIONS:

- Affected party must use commercially reasonable efforts to mitigate
- Alternative sourcing required within 60 days for delayed equipment
- Alternative shipping routes must be explored (max 20% cost increase acceptable)
- Partial performance required where possible

CONSEQUENCES OF FORCE MAJEURE:

- Time extension granted equal to FM duration plus reasonable recovery period
- No liquidated damages assessed during proven FM period
- Owner bears cost of FM event delays; Contractor bears mitigation costs
- Termination right: If FM exceeds 180 days either party may terminate with 30 days notice

KNOWN CRITICAL VULNERABILITIES:

1. SINGLE SOURCE DEPENDENCY: ASML is the ONLY supplier of EUV lithography machines globally - no alternative exists. Any disruption to ASML or Dutch export license = project failure.

2. TAIWAN STRAIT CHOKEPOINT: 100% of equipment must transit Taiwan Strait. Current geopolitical tensions between China and Taiwan. US-China strategic competition over Taiwan. Recent Chinese military drills and "blockade exercises" near Taiwan. Any Taiwan Strait closure = complete project stoppage.

3. TECHNOLOGY EXPORT CONTROLS: All equipment subject to US Export Administration Regulations (EAR). ASML machines require Dutch government export license (renewed annually). US CHIPS Act restrictions. Semiconductor technology classified as "Foundational Technology" under US-China tech competition. License denial = project impossible.

4. GEOPOLITICAL FLASHPOINT: Taiwan designated as highest-risk geopolitical flashpoint by global intelligence agencies. US-China strategic competition intensifying. Taiwan presidential elections affecting cross-strait relations. Potential for "reunification" pressure from China. Military tension in South China Sea affecting regional shipping.

5. NATURAL DISASTER EXPOSURE: Taiwan located on Pacific Ring of Fire - highest earthquake and typhoon risk zone globally. Major earthquake risk (magnitude 6-7 probable during 36-month construction). Typhoon season June-November annually. Historical precedent: 1999 Chi-Chi earthquake (magnitude 7.3) caused nationwide power outages and infrastructure damage.

6. CYBER WARFARE RISK: Semiconductor fabs are prime cyber attack targets - highest value infrastructure. Taiwan subject to continuous cyber attacks from state and non-state actors. Critical SCADA systems vulnerable. Ransomware risk. Supply chain cyber infiltration. Stuxnet-style attacks possible on centrifuge-like precision equipment.

7. ENERGY SECURITY: Taiwan imports 98% of energy needs - vulnerable to blockade or sanctions. Electrical grid operates near capacity - rolling blackouts possible. Nuclear phase-out policy increasing reliance on imported LNG. China could pressure energy suppliers. Fab requires 500 MW continuous power - any interruption damages precision equipment.

8. WATER SCARCITY: Fab requires massive water supply (100,000 m³/day). Taiwan experiencing recurring droughts (2021 worst drought in 56 years). Climate change intensifying water stress. Competing demands from agriculture and residential

use. Government water rationing possible during drought.

9. SUPPLY CHAIN COMPLEXITY: 8+ countries, 20+ critical suppliers, 15+ shipping routes, multiple chokepoints (Taiwan Strait, Suez Canal, South China Sea). Any single point of failure cascades to project delays. Lead times for equipment: 18-24 months - cannot be compressed.

10. STRATEGIC TARGET: Project designated "Critical National Infrastructure" by Taiwan government. High-value target for economic coercion. Potential leverage point in geopolitical negotiations. Subject to national security review. May be caught in US-China strategic competition over semiconductor supremacy.

INSURANCE COVERAGE:

- Comprehensive General Liability: USD 100M
- Contractor's All Risk (CAR): USD 1,500M
- Marine Cargo Insurance: USD 1,000M
- Delay in Start-Up (DSU): USD 200M
- GAPS: No terrorism insurance, No political risk insurance, No war risk insurance, No cyber insurance mandated

EXPORT COMPLIANCE REQUIREMENTS:

- US Export Administration Regulations (EAR) compliance required
- Dutch export license for ASML machines (subject to geopolitical review)
- Wassenaar Arrangement controls on dual-use technology
- Taiwan Technology Protection Act compliance
- Immediate notification required if export licenses suspended or denied

TERMINATION CONDITIONS:

- Material breach with 30-day cure period
- Insolvency or bankruptcy
- Force Majeure exceeding 180 days
- Government prohibition or export control denial
- Owner termination for convenience: 90 days notice, costs plus 10% profit

GOVERNING LAW AND DISPUTES:

- Taiwan Contract Law governs interpretation
- ICC Arbitration in Singapore - 3 arbitrators - English language
- Executive negotiation: 30 days
- Optional mediation in Singapore: 30 days
- Binding arbitration as final resolution

PROJECT SUCCESS FACTORS (ALL REQUIRED):

- ✓ Taiwan Strait remains open for commercial shipping
- ✓ No US/Dutch export restrictions on EUV technology to Taiwan
- ✓ Continued supply of rare gases from European suppliers
- ✓ Stable Taiwan electrical grid (no sustained blackouts)
- ✓ No major earthquake >6.5 magnitude during construction
- ✓ Continued operation of Suez Canal for European shipments
- ✓ No pandemic-related factory shutdowns in Netherlands/Japan/Korea
- ✓ No Chinese "reunification" military action
- ✓ Stable US-Taiwan relations and defense commitment
- ✓ No escalation of US-China technology decoupling

RISK ASSESSMENT:

This project faces EXTREME geopolitical, supply chain, natural disaster, and technology export control risks. The Taiwan Strait dependency creates a single point of catastrophic failure. The Force Majeure clause has SIGNIFICANT GAPS including terrorism, cyber warfare, trade restrictions, and supply chain disruptions. Expected FM invocation probability: HIGH (60-85%). Expected loss exposure significantly exceeds contract value due to equipment replacement costs, remobilization, and market opportunity costs for lost semiconductor capacity.

